REPUBLIC OF KENYA

STATE DEPARTMENT OF AGRICULTURE

Understanding & Managing
Maize Lethal Necrosis Disease (MLND)
in Kenya

A Training Manual for extension staff and other service providers

Preface

Development of this training manual has been necessitated by the need to train field extension officers and other service providers on the Maize Lethal Necrosis Disease (MLND). The disease has devastating effects on maize production where it occurs. It has been shown to cause up to 100% crop loss. This magnitude of loss demands that serious measures be taken to mitigate against the impact of the disease. The manual contains valuable information on the MLND with special reference to its recognition and management. It is envisaged to be a useful tool in curbing the disease, particularly through agricultural extension service providers who commonly interact and share information with farmers and other stakeholders.

This training manual has been developed by a consortium drawn from the following institutions under the guidance of the State Department of Agriculture. The institutions include:

- Plant Protection Services Division (PPSD)
- Kenya Agricultural Research Institute (KARI)
- University of Nairobi (UON)
- Kenya Plant Health Inspectorate Service (KEPHIS)
- Pest control Products Board (PCPB)
- International Centre of Insect Physiology & Ecology (*icipe*)

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ABBREVIATIONS

FAO	Food and Agriculture Organization
Icipe	International Centre of Insect Physiology & Ecology
KAPAP	Kenya Agricultural Productivity & Agribusiness Project
KARI	Kenya Agricultural Research Station
KEPHIS	Kenya Plant Health Inspectorate Service
PCPB	Pest control Products Board
PPSD	Plant Protection Services Division
MDTT	Multi Disciplinary Technical Team
UON	University of Nairobi

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PART ONE

BACKGROUND INFORMATION ON MAIZE LETHAL NECROSIS DISEASE (MLND):

Topic One: History of Maize Lethal Necrosis Disease (MLND)

Learning Objective: To enable participants understand the background information on MLND.

Procedure: Theory lesson and discussions.

Duration: 30 Minutes

a) **Importance of MLND globally**

The Maize lethal necrosis disease involves synergism between two viruses namely Maize Chlorotic Mottle Virus (MCMV), and any of several potyviruses including Maize Dwarf Mosaic Virus (MDMV), Wheat Streak Mosaic Virus (WSMV) and Sugarcane Mosaic Virus (SCMV). Maize Chlorotic Mottle Virus was first reported in Peru in 1973 (Hebert and Castillo, 1973) where it was found to cause losses of 10 and 15% in floury and sweet corn varieties. The virus is geographically restricted and is associated with soil. In experimental plots, inoculated plant yields were reduced by up to 59% (Castillo-Loayza, 1977).

In Kansas crop losses due to Corn Lethal Necrosis (synonym for MLND) were estimated to be between 50% to 90% (Uyemoto et al., 1980; Niblett and Claflin, 1978) depending on the variety of maize and the season of the year. The disease has also been reported in China. Maize Lethal Necrosis Disease poses a major challenge to maize production in regions where it has been reported.

b) **History of MLND in Kenya**

In Kenya the first occurrence of unfamiliar maize disease was first reported in September 2011 in the low altitude zones of Bomet County (Longisa Division, 1900.m.a.s.l.) affecting 200 Ha of the second season maize crop. The effects of the disease were sudden; devastating and could not be explained. The cause of the problem was unknown. Local farmers called it ‘Koroito’, which is the vernacular name for plague. By the end of 2011 the disease had spread to the higher altitudes of Bomet (2100m a.s.l), the neighboring Chepalungu, Narok South, Narok North and Naivasha sub-counties. In 2012 the disease was reported in all maize growing regions in the country except North Eastern.

c) Disease identification

As soon as the disease was reported joint collaborative field surveys and consultative meetings were held by relevant stakeholders. Samples were collected and send to different laboratories for analysis and identification of the viruses. Ohio State University USA and identified and Food and Environment Research Agency (FERA), UK identified Maize Chlorotic Mottle Virus (MCMV) and Sugarcane Mosaic virus (SCMV), respectively.

Various insects which include root worm beetles, thrips, aphids and leafhoppers have been implicated in the transmission of MCMV and SCMV. Other laboratory analysis showed presence of fungal pathogens (*Cephalosporium acremonium*) associated with the affected maize.

Maize is reported to be the primary host of the MLND viruses. However, a wide range of plants in the grass family (73 species) could be alternate host for viruses. The disease affects all varieties of maize and there is evidence of low levels of seed transmission.

d) Preparation of awareness materials

Upon identification of the disease, the technical materials such brochures and posters for use by Extension staff and farmers were prepared and this training manual has been developed to guide in training of extension service providers.

e) Sensitization of stakeholders

Field days were conducted to sensitize farmers in the maize growing areas where the disease was first reported. Radio programmes and TV advertisements were also prepared and aired in the local media.

f) Resource mobilization

The technical team held several consultative forums to share information and develop a research proposal for sustainable management of MLND in Kenya. The proposal was submitted to funding agencies such as FAO and World Bank through Kenya Agricultural Productivity and Agribusiness Project (KAPAP). The funding agencies have agreed to support various activities within the proposal to generate knowledge and information that provide an understanding of the disease, its vectors and interactions. This knowledge will help to design informed management options for the disease.

g) Constraints in Maize farming in Kenya

Maize is a very important food security crop as well as cash crop for majority of farmers. However its production faces a number of challenges. Some of these challenges include:

a) Erratic rainfall - poor distribution of rainfall results in total or partial crop failures,

b) Pests such as Stalk borer causes damage ranging from 60 - 95% yet many farmers take no control measures,

c) Inadequate skills in diseases monitoring and pests scouting by farmers. Only 20% of farmers surveyed actual undertake regular scouting for pests and diseases this makes it more difficult to execute prevention and eradication measures at the right time.

d) Crops damage by wildlife was also raised by farmers that neighbor forested areas.

e) Diseases especially those with significant damage on leaves (Chlorosis and necrosis), stem (rot and lodging) and cobs (rotten and moldy).

f) Soil pests (termites, nematodes, cutworm, chaffer grubs), Rodents and Birds damage especially in maize from planting to cob formation,

g) Nutritional deficiency symptoms i.e. phosphorus especially in lower Eastern region were high,

h) Inadequate appropriate storage capacity at farm level, Lack of appropriate storage capacity poor post harvest management at small scale farm level results in heavy losses to harvested grains.

i) Continuous land subdivision into uneconomical units is affecting seed maize and commercial production.

j) Competition from cash crops such as sugarcane, tea, wheat, etc in major maize growing zones is reducing hectare available.

h) Questions and Answer session

- When the disease was first reported in the world and in which country?
- Name the disease causing orgasms in MLND complex
- What is the name of the district and division where the disease was first reported in Kenya?
- What are the potential insect vectors closely associated with the disease in Kenya?

CHAPTER TWO: **MLND DISEASE SYMPTOMS AND MANAGEMENT.**

Topic Two: Symptoms of MLND.

Training objective: To enable participants to identify the disease symptoms.

Procedure: Lecture, slide show of disease symptoms, visit to the field and group discussions.

Duration: 45 minutes,

a) General Field Expression of MLND Symptoms

Infected plants show a diverse range of symptoms. The can be observed on all above ground parts of the plant (Tassels, leaves, Stem and Cobs). The symptoms are observed on plants as early as 2 weeks after germination up to cob formation stage.

I. Leaf symptoms

These include Chlorotic mottle on the leaves, usually starting from the base of the young leaves in the whorl and extending upwards toward the leaf tips **(Fig 1a),** mild to severe leaf mottling, dwarfing and premature aging **(Fig 1b).** Necrosis of young leaves in the whorl before expansion leading to a 'dead heart' symptom **(Fig 1c**



Fig 1a Fig 1b Fig 1c


Diffused and dot like symptoms





Severe mosaic, chlorosis, streaking

Diffused yellowing and necrosis



Mosaic with advanced marginal and tip leaf necrosis and Dead Heart Symptom.

II. Stem symptoms

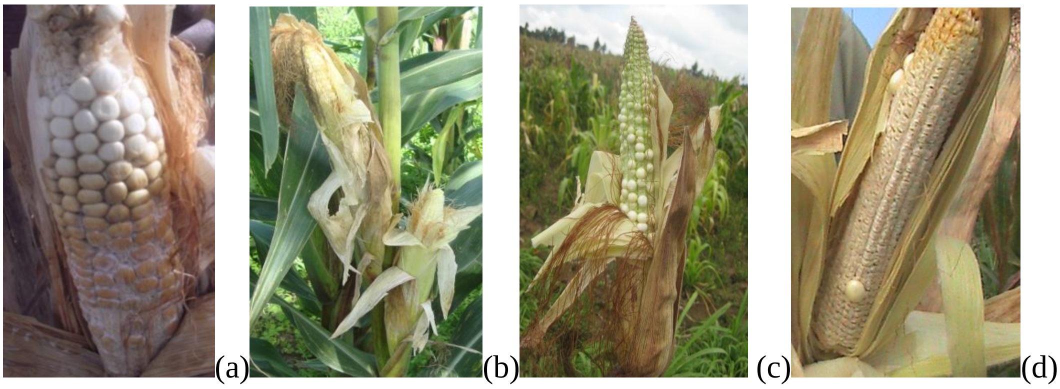
Lesions sometimes associated with unfurled leaves appears on the stem.

III. Tassel Symptoms

Poorly filled Kernel and Tassels with no pollen

IV. Cob and Grains Symptoms.

Later during crop growth and development, fungal symptoms are also observed. They include Chlorosis, necrosis with leaf reddening, discoloration of internodes, brownish white moldy growth on rotting cobs **(Fig 2a)**. Ear husks dry when the rest of plant is green **(Fig 2b**). Partial grain or no grain filling **(Fig 2c & d)** and all infected plants may die giving blighted appearance of the crop. Insects observed on the infected plants include thrips, aphids and stalk borers.



(a) (b) (c) (d)

Fig 2: Symptoms on affected cobs.

b) Question and Answer session

- At what stage/age does the MLN symptoms start showing on maize plant?
- Which parts of the plant express the MLN symptoms first?
- What is the appearance of the cobs affected by the Disease?

c) Group Discussions

Provide slides of MLN disease symptoms and symptoms caused by biotic and abiotic factors for identification.

Topic Three: Disease transmission and spread.

Learning Objective: To enable the participants understand the transmission and spread of MLN disease in the maize fields

Procedure: Lecture, field observations and group discussions.

Duration: 45 minutes

a) Disease transmission and spread

The transmission of MCMV can be mechanical, by several insect vectors (maize thrips, Aphids, maize rootworms beetles, leaf beetles and leaf hoppers) and by seeds at very low rates (Jensen et al. 1991). The virus pathogen (MCMV) can also be transmitted through soil and through infected plant debris since the virus can survive in plant residues (Nyvall, 1999). Continuous maize production in a field greatly increases the incidence of the viruses and vectors.

Although maize is the preferred host of corn thrips they can also survive on a number of plants, including cassava, beans, maize, sorghum, onions, various grasses, rice, peppers coriander, peas, and the weedy species *Bidens pilosa* and *Tithonia diversifolia* (Capinera 2008; ICIPE Thrips, 2011; Frison and Feliu 1989; King and Saunders 1984).

Insects that have been shown not to transmit MCMV include several aphid species, two leafhoppers, a planthopper, a whitefly, a scarab beetle and a noctuid larva. However, aphids, planthoppers and whiteflies are able to spread other viruses. Aphids spread some of the potyviruses (MDMV& SCMV) that interact with MCMV to cause MLND. Planthoppers spread Maize streak Virus (MSV) which also affects maize and can interact with MCMV. Other pests believed to transmit MCMV include *Peregrinus maidis, Sardia pluto, Empoasca solana, Adoretus sinicus*, *Tetranychus sp*.

b) Question and answer Session

- Which pests are suspected to be the key vectors of MCMV virus
- Which pests are suspected to be the key vectors of SCMV virus
- Name at least 4 main vectors responsible for the transmission of the disease.

Group work: In a slide show or field show other host plants and potential vectors

Topic Four: Management of MLND

Learning Objective: To enable participants understand and train others on MLND management,

Procedure: Lecture, slide shows, demonstrations, group discussions

Duration: 1 hour

a) Management of MLN Disease in Maize

The most effective control for MCMV has been achieved through the integration of cultural practices, control of vectors and host resistance (Nelson et al., 2011). A number of sources of resistance to MCMV have been identified and are being incorporated into commercial maize varieties throughout the Western Hemisphere (Nelson et al. 2011). Alternatively, crop rotation with non-maize (Graminae) crop has been shown to reduce the incidence of MCMV the following year (Phillips et al., 1982; Uyemoto, 1983). Reports indicate that many temperate inbred and hybrids are highly susceptible to the disease, while resistant tropical germplasm has been developed (Nelson et al., 2011).

I. Insect Vector Control

Thrips are small insects, about 0.04 inches long. Adult thrips have two pairs of narrow wings which are fringed with hairs. Immature thrips are wingless, whitish to yellowish in color, and are most commonly found in whorls, tassels, ears, or on the underside of leaves. Adults emerge continuously throughout the warm months. Adults and immature may be found in maize at any time during the growing season. Eggs are deposited in plant tissue and hatching occurs in about 5 days; the immature stages take about 5 to 7 days to complete development. Thrips population increase in warm dry periods thereby increasing the activity of the insects and the spread of the disease. Vector control can be achieved through seed treatment that covers the plants at the seedling stage and application of foliar sprays with systemic pesticides from three weeks**.**

a. Management Strategies for the vectors

Chemical control is warranted only if the natural enemies and parasites of the maize leaf aphids are not present and aphid densities are above the threshold population. There are several natural enemies that exist and are quite effective at controlling aphids. These include ladybird beetle adults and larvae, lacewing adults and larvae, and a few parasitic wasps. Chemical control will kill natural enemies and may lead to a resurgence of the aphid population. Spray with recommended pesticides only.

II. Cultural Control

a. Field Hygiene.

Field hygiene, closed season, promotion of GAP, crop rotation and are the most important methods of control of MLND in Maize. Thrips populations tend to build up on weeds. It's important to cultivate nearby weedy areas before maize emerges to reduce the potential of thrips build up. Cultivating weedy areas after maize emergence will increase thrips problems as they will migrate from the weeds to maize plants nearby. You should also undertake regular surveys of the maize plots and remove immediately any affected plants. During this exercise care must be taken not to spread the disease by putting all the affected plants in polythene paper or gunny bags before moving the material. All affected vegetative materials should be either burned or given to livestock.

b. Design and enforce maize closed season in all regions,

Farmers in some parts of the country practice continuous maize production. This creates a conducive environment for the spread of the disease. We need to design and implement a closed season programme by documenting planting calendar for each area. All farmers should plant at the prescribed time of the year only. Beyond this period the farmers should be encouraged to plant alternative crops to maize.

c. Design and implement Crop Rotation Schedules for all regions

Most areas of these country farmers have been growing a wide range of crops. Some of this crops are more adapted and their returns is higher compared to maize but have stopped planting them for various reasons. Identify the constraints and address them then promote the crops once again.

III. Regulatory control

Impose quarantine on movement of maize materials from affected areas to other areas. This may be complemented by carrying out public awareness campaigns to sensitize on the effects of the disease.

IV. Good Agricultural Practices (GAP)

Farmers are required to plant only certified treated seeds with appropriate insecticide and avoid use of recycled seeds. This is the normal practice especially for open pollinated varieties. Plant early and apply adequate soil amendments including manure, basal, foliar and top dressing fertilizers to boost plant vigor and hence tolerance to pests and diseases. Enhance crop diversification to improve soil fertility and reduce risks caused by overreliance to one crop.

V. Plant Host resistance

This is the most effective control options but will require a minimum of four years. It will involve screening of commercial maize varieties and lines for tolerance/resistance to MLND and Transgenics i.e. inclusion of MLND tolerance/resistance genes in commercial maize seeds.

VI Biological control

This is the use of the entomo-pathogens that are available for control of insects particularly thrips, aphids and the plant bugs. These include *Bacillus thruingiensis*, *Metarhizium anisoplae* and *Beauvaria bassiana.*

VI. Monitoring and Response for MLND

Outbreaks of emerging pests and disease have become quite regular due to climate change and environmental degradation. It strongly recommended that each county to establish a Rapid Response unit with representation from relevant disciplines to;

- Monitor similar outbreaks using sticky and coloured traps (yellow and blue) to indicate the presence and the populations present for planning and to take appropriate action in time.
- Undertake emergency response as necessary,
- Execute containment & eradication measures,

- Collect data document and share with all concerned stakeholders,

b) Question and answer session

- Names various control options for MLN disease?
- Which are the most effective methods of control for the MLN disease?
- What different methods can one use in reducing the population of insect vector in maize fields

c) Group Work

Discussions on the spread of the disease

Identification of alternative crops, Disease distribution, symptoms, agronomic practices, presence of insect vectors,

List or identify the alternative crops to maize for different regions in the country and their production challenges.

PART II

1. OTHER PATHOGENS ASSOCIATED WITH MLND

Topic Five: Fungi and Bacterial pathogens associated with MLND

Learning Objective: To enable the participants to:

i) gain knowledge on other pathogens associated with MLND

ii) describe the symptoms and identify the pathogens

iii) discuss methods of disease management

Procedure: Lecture, Slide shows, field observations, group discussions.

Duration: 1 hr

Introduction

The maize plant is susceptible to many diseases caused by infectious and non-infectious causal agents. The diseases cause yield and quality loss of maize. The purpose of this section is to help with recognition of the common diseases of maize in Kenya caused by infectious pathogens such as fungi, bacteria, viruses and nematodes. Accurate disease diagnosis helps in the understanding of when, where the disease occurs and how it can be managed based on plant symptom description. The disease symptoms are observed at different stages of maize development; Stage I: seedling emergence to knee high; Stage II: knee high to tasselling and Stage III : tasselling to maturity

a) Fungal pathogens

Northern Corn (Turcicum) Leaf Blight (NLB) caused by (*Exhelosporium turcicum*) is favoured by cool, wet weather and is aggravated by previous diseased leaf tissue left on the soil surface. The disease is observable during the II and III stage of maize development. Early symptoms include chlorotic “halo’ that develop into necrotic lesion. These grow into mature cigar-shaped lesions that are 2 cm wide and about 15 cm long. Immature lesions have gray green margins and as they mature turn brown with dark parts of fungal sporulation. The symptoms appear first on lower leaves and increase in size and number as the plant develops, until foliage is completely “burned”. The disease can be managed by:

j) Ploughing in or removal of the diseased leaf materials from the soil surface

ii) Planting resistant varieties, iii) crop rotation and iv) spraying with fungicides before the disease becomes severe (fig 1 a & b)

Fig 1 a & b Cigar shaped necrotic lesion against a background of green color.

Gray leaf spot (*Cercospora zeae-maydis, C. sorghi var. maydis*) GLS

This disease, also known as Cercospora leaf spot, develops in highly humid warm, overcast days. Lesions begin as small, regular, elongated brown-gray necrotic spots growing parallel to the veins. Mature lesions are oval in shape and occasionally, lesions may reach 5cm long and 0.3 cm wide. The lesions start from the lower leaves and increase in number usually after 'silking' in stage III of plant development. They are always light and develop into gray lesions which may grow together and kill entire leaves. Minimum tillage practices have been associated with an increased incidence of GLS.

Management: plough under diseased debris to reduce amount of inoculum, plant resistant varieties, practice crop rotation and spray with fungicides

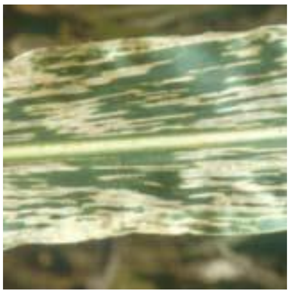
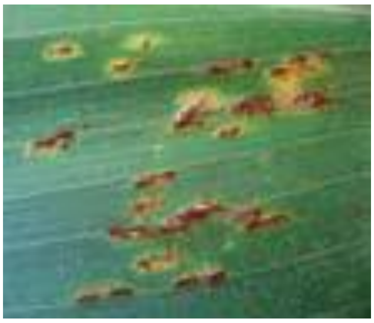


Fig Gray leaf spots on maize leaves restricted along the veins a) immature lesions looking silvery b & c older lesions that are brown and darkened cause of sporulation

Common rust (*Puccinia sorgi)*

The disease is characterized by elongate raised bumps (pustules) scattered or clustered together on both leaf surfaces that are red to dark brown in color. The symptoms are mainly observed in the mid and upper canopy of the crop in stage II and III of crop development. These occur most conspicuously when plants approach tasseling. Later the epidermis is ruptured and the lesions turn black and spores are released as the plant matures. The alternate host Wood sorrel (Oxalis spp.) is frequently infected with light orange colored pustules another stage of the same fungus. Disease spread is favored by humid warm conditions and in severe cases cause death of the leaves. Management: grow resistant varieties, control alternate hosts and spray with fungicides


a

b

c

Fig a & b **Dark brown pustules on maize leaf surfaces and C) another stage of the same fungus on Oxalis spp**

Common smut (*Ustilago maydis*)

The disease occurs mainly in the humid and temperate areas as opposed to hot humid tropical areas. The pathogen survives in soils as teliospores in a wide range of temperatures (16-35oC). The fungus attacks all parts of the plant that is leaves, stalks, tassels and ears even below the soil surface. It gains entry through wounds or thin walled cells of actively growing maize. Common smut is characterized by galls that replace individual kernels and are covered with white membranes. In time the galls break open releasing black masses of spores that will infect maize in the following season. It is different from head smut for lack of the vascular bundles that appear as fiber in smut -infected ears. On the leaves the disease appears as white blisters that turn dark with age. The blisters tend to harden and dry without rupturing. The disease is most severe in young plants and can stunt or kill them. Management: Burn affected plants completely, plant resistant varieties, spray with fungicides, and avoid injuring the plants (Fig)



Fig a & b Affected ears showing galls covered with white membrane and others that are broken and are revealing the black masses C) leaf with harden and dry blisters resulting from common smut infection

Head Smut (*Sphacelotheca reiliana*)

Head smut can cause significant economic damage in dry, hot maize growing areas, and in mid hill zones and under warm temperate conditions. The infection is systemic: the fungus penetrates the seedlings and grows inside the plant without showing symptoms, until the tasseling and silking in stage II and III of crop development. The most conspicuous symptoms are:

- Abnormal development of the tassels, which become malformed and overgrown,
- Black masses of spores that develop inside individual male florets; and
- Masses of black spores replace of the normal ear, leaving the vascular bundles exposed and shredded looking like fibers
- Plants may be severely stunted


a

b

c

Fig a& b deformed tassels and ears and C) the male florets replaced with the black powdery masses

Fungal Ear and Kernel rots

These rots can affect ears, kernels, or cobs, reducing test weight and grain quality. Some rots are responsible for development of mycotoxins that may contaminate grain. Positive identification is difficult. Rotting observed in the field is often due to a complex of causal organisms, not just one. Most ear rots are favored by late-season humidity. Infections are increased by ear damage by birds or insects and by stalk lodging that allows ears to come into contact with the soil.

Aspergillus Ear and Kernel Rot *(Aspergillus flavus, Aspergillus spp)*

The disease may be a serious problem when infected ears are stored at high moisture content. Several species of Aspergillus can infect maize in the field. *Aspergillus niger* is the most common; it produces black, powdery masses of spores that cover both kernels and cob. Greenish or yellowish-tan discoloration occurs on and between kernels, especially near the ear tip. Symptoms are more prevalent if the husk does not cover the ear tip in stage II of crop development. The rot is favored by hot, dry weather. It may produce aflatoxins.


Ear and kernel rot by Aspergillus

Diplodia (Stenocarpella) Ear Rot)

Symptoms include bleached husks, white mold over kernels, and rotted ears with tightly adhering husks. Early infection (2 to 4 weeks after silking) is likely to lead to complete ear rotting. Later infections may result in partial rotting, usually beginning at the base. Since corn is the only known host, this disease is most severe when maize is planted following corn in reduced tillage situations.

 White mould of Diplodia spp on rotten ear

Fusarium Kernel or Ear Rot *Fusarium moniliforme*

In contrast to the damage from *G. zeae*, damage from *F. moniliforme* occurs mainly on individual kernels or on limited areas of the ear. Infected kernels develop a cottony growth or may develop white streaks on the pericarp and germinate on the cob. Scattered individual or groups of kernels show whitish-pink to lavender fungal growth. Infected kernels may also have a "starburst" pattern of white streaks on the cap of the kernel or along the base. Infections are more frequent on damaged ear tips, and are favored by dry weather. Fusarium rot may produce mycotoxins known as fumonisins, which are harmful to several animal species.

 a  b

Fusarium infected ears

Gibberella Ear Rot *(Giberella zeae***)**

Gibberella zeae, the sexual stage of the pathogen, is most common in cool and humid weather, particularly 2 to 3 weeks after silking. The fungus produces mycotoxins known as deoxynivalenol, zearalenone, and zearalenol, which are noxious to several animal species. Symptoms include reddish kernel discoloration, usually beginning at the ear tip. Ear infection begins as white mycelium moving down from the tip, which later turns reddish-pink, in infected kernels. Husks may rot and be cemented to the ear.

 Giberella infected ear causing red colouration on kernels

Photo Citations:

Clemson University - USDA Cooperative Extension Slide Series, Bugwood.org (viewed 9/22/10); Corn Stunt 1235014

William M. Brown Jr., Bugwood.org (viewed 9//22/10) High Plains Virus 5366657

Howard F. Schwartz, Colorado State University, Bugwood. Org (viewed 9/22/10) Bacterial Stalk Rot 5361254

Holcus spot and Pythium stalk rot photos courtesy of Don White at University of Illinois

Bacterial pathogens

I. **Stewart's wilt** *(Erwinia stewartii, syn. Pantoea stewartii)*

The bacterial pathogen is reportedly transmitted by maize flea beetles (*Chaetocnema pulicaria*) and also at very low frequency through infected seed. With early infection, lesions are first water-soaked, turning long and pale yellow with irregular margins running the length of the leaves. Infection may move into the stem, causing a general stunting, wilting, and plant death. Severely infected plants that set seed develop small nubbins with few kernels. Late infection can cause severe leaf necrosis but does not lead to wilting. Feeding wounds from the insect vector serve as points of entry for the pathogen, which is carried from one season to the next by the flea beetle. Please note that this disease has not yet been reported in Kenya and is therefore of quarantine importance.

Maize plants affected by the bacteria (*Erwinia stewartii*)

k) Management of fungal and bacterial Diseases

a. Cultural

Practice crop rotation to reduce the buildup of insect vectors and the disease inoculums. Farmers should be encouraged to submit samples of affected plants for laboratory analysis.

b. Chemical

Control the vector and also the disease using the recommended pesticides by Pesticide Products Control Board.

l) Question and answer session

- What are some of the important bacteria that cause economic loss in maize crop?
- What are some of the important fungi that cause economic loss in maize crop?
- What is the most effective method of control of bacteria disease in maize
- What is the most effective method of control of bacteria disease in maize

Important viral pathogens that affect maize

Maize streak virus

The disease, reported first from East Africa, has now extended to other several African countries. The virus is transmitted by *Cicadulina* spp. leafhoppers. *Cicadulina mbila* (Naude) is the most prevalent vector, and will transmit the virus for most of its life after feeding on an infected plant. Early disease symptoms begin within a week after infection and consist of very small, round, scattered spots in the youngest leaves. The number of spots increases with plant growth; they enlarge parallel to the leaf veins. Soon spots become more profuse at leaf bases and are particularly conspicuous in the youngest leaves. Fully elongated leaves develop a chlorosis with broken yellow streaks along the veins, contrasting with the dark green color of normal foliage (Photos 97, 98). Severe infection causes stunting, and plants can die prematurely or are barren. Many cereal crops and wild grasses serve as reservoirs of the virus and the vectors.


a


b

a) Maize plant with streak symptoms from MSV b) a close up of the streaks on the leaves parallel to the veins

Maize Dwarf Mosaic Virus

These viruses are transmitted by several genera and species of aphids, including *Rhopalosiphum maidis* (Fitch) and also by seed at low rates (Photo 89). After feeding on an infected plant, the aphid can immediately transmit the virus. These pathogens can infect other grass and cereal hosts, such as sorghum, Johnson grass, and sugarcane. No infection occurs in broad-leaf plants. Infected plants develop a distinct mosaic— irregularities in the distribution of normal green color—on the youngest leaf bases (Photo 90). Sometimes the mosaic appearance is enhanced by narrow chlorotic streaks extending parallel to the veins. Later on, the youngest leaves show a general chlorosis, and streaks are larger and more abundant (Photo 91). As plants approach maturity, the foliage can turn purple or purple-red. Depending on time of infection, there may be severe stunting of the plant. Plants infected early may produce nubbins or be totally barren.

In China, SCMV has been reported as seriously affecting maize production.

 a  b

a. Irregular distribution of the normal green color causing a mosaic enhanced by narrow chlorotic streaks. b) Enlarged and more abundant streaks

m) Question and answer session

- What are some of the important viruses that cause economic loss in maize crop?
- What is the effect on the plant when the viruses exist as a co- infection in the plant?
- What is the most effective method of control of viral diseases in maize?

2. OTHER MAIZE PESTS & DISEASES

Topic Six: Factors affecting maize production in Kenya,

Learning Objective: To gain knowledge on other abiotic factors that contributes to MLND expression in maize field,

Procedure: Lecture, slide shows

Duration: 1 hr

a) Field Pests

Insect pests severely limit the production of maize, one of the most important cereal crops worldwide. Losses reach millions of dollars annually. Many factors limit maize production; insects and mites being among the most important. Insects that infest maize are chaufer grubs, stem borers, armyworms, ear borers/ African bollworm, leaf hoppers, thrips and aphids. The key to an insect's success lies in its great reproductive potential, small size, efficient dispersal mechanisms, and ability to survive harsh environments. Some of these insects have been reported to potentially transmit viral diseases in maize from other parts of the world (refs). They include:

Larvae of scarab beetles (White grubs)

The larvae/grubs of Scarab beetles are white in color. The hind part of the body is smooth and shiny with dark body contents showing through the skin. Grubs are slightly longer than an inch when fully grown and curl into a “C” shape more often. The grubs have a 3 year life cycle. They eat roots and severely cause wilting in which the center leaves of the young maize plant die. During the feeding, the grubs are reported to transmit MCMV that is acquired from plant debris on soil surfaces when maize is off season.

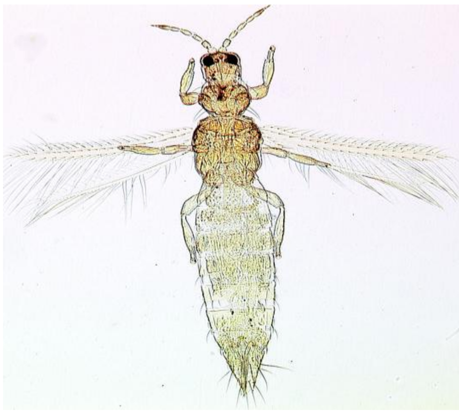

a. A fully grown white grub/ larvae of scarab beetle

Maize thrips (*Frankliniella williamsi*)

Grass thrips found in maize are minute (about 1/16th inch) elongate, yellow insects. The adults have slender wings fringed with long hair. The immature are wingless and yellow orange. The thrips damage the plants by rasping the leaf tissue and sucking the exudates (plant sap) from the ruptured cells. They feed deep in whorls under the leaf sheath and on the underside of the lower leaves. When injury is extensive, and the maize plants are stressed, leaves and ear husks may appear to have a silvery cast. Thrip feeding is never an economic problem unless the maize plants are young less than one foot tall**.**

Thrips are most noticeable and of greatest concern at two periods during the maize growing season; on young seedling plants and at ear formation. On young seedlings their feeding makes the plants wilt and can cause stunting. A common sign of a heavy thrips infestation is distorted leaves that turn brownish around the edges and cup upwards. Despite infestation by the thrips the plants are able to tolerate the attack. At ear formation, mechanical damage by thrips to developing kernels provides entry by Fusarium spp. causing Fusarium ear rot diseases. The maize thrips are reported to spread MCMV which in combination with any potyvirus (SCMV or MDMV) causes MLND.

a


b

a. Larvae of maize thrips and b) adult thrip

Maize aphid (Rhopalosiphum maidis)

Adult aphids are either winged or wingless less than 1/25th inch long. They are blue - green with black antennae, legs and tail pipe (cornicles). The nymphs look similar to adults but are smaller, light green and wingless. The soft bodied insects colonize the tassels and the upper leaves of the maize plants. Their feeding removes moisture and nutrients from the plant and excess sap is secreted as honey dew and may cover the tassels and the leaves. Colonies of aphids found in central leaf whorl. The insect feed on the inflorescence and young leaves especially during dry periods causing yellow mottling of leaves. The aphids are reported to transmit barley yellow dwarf virus (BYDV) and a vector of MDMV (ref).

 a  b  c

a) Wingless aphids on maize leaf b) a mass of aphids on the tassel stem c) a mass of insects on leaves

Leafhoppers

b) Nutritional deficiencies of maize

Learning Objectives: Participants to able to describe the importance of nutritional disorders of maize; discuss basic aspects of nutritional disorders of maize & differentiate between nutritional deficiencies and disease symptoms.

Procedure: Lectures, Slide show, discussions

Duration: 30 minutes

Introduction

All essential elements are by definition required for plant growth and completion of the plant life cycle from seed to seed. Some essential elements are needed in large quantities and others in much smaller quantities. However, from a practical standpoint, three of the six essential macronutrients are most often "managed" by the addition of fertilizers to soils, while the others are most often found in sufficient quantities in most soils and no soil amendments are required to supply adequate supplies.

The **primary nutrients** include N, P, and K, because they most often limit crop production. All of the other essential macronutrient elements are **secondary nutrients** because they are rarely limiting, and more rarely added to soils as fertilizers.

The ability of soils to supply secondary nutrients to plants indefinitely is subject to the law of conservation of matter and is therefore dependent upon nutrient cycling. Continued crop removal of Ca, Mg, and S requires replenishment just as surely as primary nutrients, but most likely less frequently. Calcium and magnesium are often supplied by mineral weathering, either of natural soil materials or of **aglime**, ground limestone added to correct soil acidity. Sulfur is often added to soil as either atmospheric deposition (associated with air pollution) or as impurities in fertilizers, particularly common P fertilizers.

To demonstrate that this classification is more responsive to soil ability to supply nutrients than plant requirements, it should be noted that plant requirements for Ca, a secondary nutrient element, is greater than for P. Calcium is found as a principle exchangeable cation in most soils and an important soluble cation in the soil solution. Phosphorus, on the other hand, is only slightly soluble in most soils, and many soils (particularly acid soils and alkaline soils) have the potential for causing phosphorus deficiencies.

Macronutrients: N, K, Ca, Mg, P, and S, and Micronutrients include: Cl, Fe, B, Mn, Zn, Cu, Mo, and Ni.

Most growth-limiting nutrient will limit growth, no matter how favorable the nutrient supply of other elements. For example, a deficiency of Fe or Mn (most common in soils containing calcium carbonate) can severely limit plant growth in spite of adequate N, P, and K.

i. Nitrogen (N) deficiency

Nitrogen (N) deficiency makes the older leaves (the bottom portion of the maize plant) turn pale or yellowish-green. The deficiency then starts to create a V shape, starting at the tip of the leaf. If not rectified, the deficiency works its way up the plant from older (lower) to newer leaves. The stalks tend to be thin and spindly. N deficiency develops commonly in wet to saturated soils or under cool soil temperatures. N can leach out under heavy rainfall in light-textured (sandy) soils or can be denitrified in flooded soils when temperatures are warm. N deficiency can be induced after midseason or during other periods when soils tend to be dry. N deficiency can also occur in soils with large amounts of low-nitrogen-containing residues.

Nitrogen Deficiency

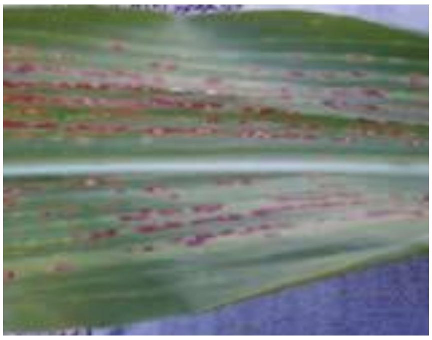
ii. Phosphorus (P) deficiency

Phosphorus (P) deficiency causes a distinct dark green with reddish to purplish leaf margins, typically starting from the tip. The deficiency is observed in the older leaves. Stunted growth is also typical. At early development stages some hybrids show purple colors even though P is not deficient, while other hybrids might not show this coloration even when P levels are limiting. P deficiency symptoms normally disappear by the time the plant is waist-high. Since P is fairly immobile in the soil, any soil condition that limits root growth (cool temperature, wet or very dry conditions, compaction) can induce the deficiency.

iii. Potassium (K) deficiency

Potassium (K) deficiency is observed as yellowing and necrosis (death) of the edge of older leaves. When the problem persists, this deficiency will continue to move up from older to newer leaves, while the top leaves may look completely green. K deficiency can cause lodging of the crop later in the season because stalks are thin and not strong. As with P, soil conditions that restrict root growth can induce deficiency, especially at early stages of development when the root system is small. Soils with low K buffer capacity can cause the deficiency if an appropriate fertilization plan is not followed.

Potassium Deficiency



Magnesium Deficiency

iv. Magnesium (Mg) deficiency

Magnesium (Mg) deficiency appears in lower leaves as yellow or white streaking between veins. The leaves eventually become reddish-purple, and the edge and tip die if the deficiency is severe. Acidic and sandy soils are most likely to be deficient in Mg.

v. Sulfur (S) deficiency

Sulfur (S) deficiency causes yellowing of the foliage. S deficiency is often confused with N deficiency. Since S is not easily translocated, deficiency tends to be more visible in the newer leaves as opposed to N deficiency which is more pronounced in older leaves. The condition is typically observed in soils with low organic matter (including sandy soils), low pH, cold and wet conditions that reduce mineralization (release) of S from organic matter. Since S is leachable, maize will often recover from an S deficiency once the root system reaches the S that has accumulated in the subsurface soil.

Sulfur Deficiency

Zinc Deficiency

vi. Zinc (Zn) deficiency

Zinc (Zn) deficiency appears between veins as light green to white stripes or as wide bands starting at the base of the leaf and extending toward the tip of the newer leaves. The edge of the leaf as well as the midrib usually stays green but in cases of severe deficiency, new leaves are almost white. Zn deficiency is most commonly observed in soils low in organic matter, sandy soils with high pH (>7.3), cool and wet soils.

Note

It is important to confirm any deficiency before trying to correct the problem. Since the visual symptoms are sometimes not clear-cut, it is advisable to collect soil samples and affected plants for nutrient and pathogen analysis.

c) Weeds affecting maize

Learning Objectives: Participants to describe the importance of weeds in maize pest interactions; the effects of weeds on yield reduction & discuss basic aspects of damage, biology, and geographical distribution of weeds;

Procedure: Lectures, samples and slide show, discussions

Duration: 45 Minutes

Introduction

Weeds often germinate before the main crop, and grow with more vigor thereby competing for nutrients, sunlight and water. Their impact is critical especially during the first 3 to 5 weeks following emergence of the crop. It is important to control weeds in a maize field before they are 15 to 20 cm high, which is when they begin to impact on yields. To obtain maximum yields, weeds must be controlled before the critical period is reached.

Crop yield losses due to weed depends on crop competitive ability, weed species and density. Based on the same parameters, maize yield losses associated with weeds range from 10-100%.

a) Common weeds of maize

Besides the parasitic weeds *Striga hermonthica* (purple witchweed) and *Striga asiatica* (red witchweed), maize is infested by a wide range weeds which include broad–leaved such as *Amaranthus species, Cleome gynandra,* Wondering jew *(Commelina benghalensis), Ageratum conyzoides,* black jack *(Bidens pilosa),* Gallant soldier *(Galinsoga parviflora),* Chinese lantern *(Nicandra physalodes), Launaea maizeuta, Sonchus oleraceus,* Mexican marigold *(Tagetes minuta), Oxygonum sinuatum, Portulaca oleracea,* Sodom apple (*Solanum nigrum),* thorn apple *(Datura stramonium), Chenopodium murale, Four leaved clover (Oxalis species).*

Maize is equally infested by grass weeds such as *Cynodon nlemfuensis, Digitaria abyssinica, Digitaria velutina, Eleusine indica, Setaria spp, Avena fatua and Dactyloctenium aegyptium.*

Pictorial samples of broad-leaved weeds

Oxygonum sinuatum

Amaranthus hybridus

Tagetes minuta (Mexican marigold)

Portulaca oleracea (Purslane)

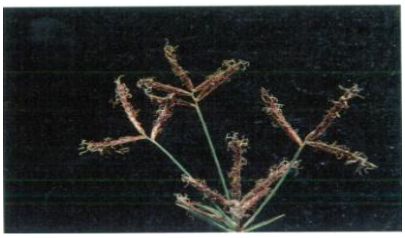
Pictorial samples of grass weeds

Eleusine indica (wild finger millet)

Setaria verticillata (love grass)

A number of sedges are also known to infest maize crop in different regions depending on habitat suitability and the most common ones include *Cyperus esculentus, Cyperus rotundus, Cyperus blysmoides, Cyperus rigidifolius* and *Cyperus teneristolon.*

Pictorial sample of Sedges



Cyperus rotundus (purple nut sedge)

b) Weed management

Weed management in maize should be based on an optimum combination of cultural, mechanical, and chemical practices.

I. Preventive methods

Prevention includes practices that minimize introduction, propagation, and spread of weeds. For example: destroying the weeds before they set seeds, planting weed-free seed, using clean equipment on the farm, and keeping the field margins clean to prevent weed invasion.

II. Quarantine

This involves exclusion of weeds from areas they have not been introduced using legislation. It restricts movement of infected/infested plant materials or noxious weeds from one area to another in the country or one country to another. It has moral obligation to protect a region or country rather than a farm.

III. Crop Rotation Practices

Crop rotation can be an important component of a weed management program. Most annual grass weeds can be more easily and/or economically managed in soybeans than in maize. The opposite is true for most annual broadleaf weeds

IV. Use of Herbicides

Herbicides are the newest and often the most efficient weed management tool. Although herbicides have become a necessity in most maize production systems, far too few growers equate weed management with herbicides.

A herbicide or herbicide combination should be selected on the basis of its effectiveness on the different weed species in the field. The correct herbicide rate must be used to obtain good weed control results and to minimize maize injury.

V. Soil weed seed bank management

Soil acts as the bank for thousands of weed seeds and the management practices applied determine the amount of weed seeds it contains since they either increase or reduce the seeds.

VI. Manual (hand weeding) method

One major disadvantage is that it is time consuming and hence in many countries, it is only done to the extent that the family labor can permit. Others are quick re-establishment in wet conditions if partially chopped or pulled out. Some weeds resist hand hoeing e.g. *Cynodon dactylon. Sorghum halepense.* Its efficient use is difficult or impossible on wet soil.

VII. Mechanical cultivation method

In small farms, peasant farms, mechanical cultivation is usually done by animal traction, rarely by tractor.

VIII. Cultural methods

Various agronomic practices are used to manage weeds. These include: Crop rotation, Conservation Agriculture and Mulching.

IX. Conservation Agriculture

Conservation agriculture (CA) is a concept for resource-saving agricultural crop production that strives to achieve acceptable profits together with high and sustained production levels while concurrently conserving the environment. CA is based on enhancing natural biological processes above and below the ground. It can be practiced in different agro ecological zones and terrain. Advantages include:

- Improved soil and water conservation,
- Enhancement of soil organic matter content (when plant material is used),
- Tropical environments high temperature of mulches ameliorates soil temperature from low to optimal.

Its disadvantages include:

- Lack of skills by farmers and service providers
- Require resources to cutting and transportation sufficient amounts of plant residue and competing uses for plant residue mulch with livestock,
- Spread of pests, diseases and weeds from one farm to another,
- Initial capital for establishment of CA is high,
- Nutrients imbalance e.g mulch rich in Ca only
- Depresses soil temperature in highland /cooler areas

FARMING BUSINESS

Strategies for Improvement of Farm productivity

"There's this belief that in order to stop poverty, we have to find ways to get people to stop being farmers. What we need to do is find ways to stop them from being poor farmers."

Learning Objective:

This guide is intended to enable Participants to:

- Perform simple calculations of costs and earnings for different enterprises in crop production
- Develop capacity to guide farmers in making informed decisions regarding the viability of alternative crops to maize
- Assist farmers access credit for farming business
- Teach farmers how establish and maintain farm records

Procedure: Lecture, Practical's and discussions

Duration: 40 Minutes

a) Introduction.

The solution to world hunger is teaching the farmers to farm profitably and sell locally. Farming is a business like any other business. It involves buying of inputs, hiring of farm labor and selling of the farm produce. For better management of the business, farmers or managers need to know how to keep simple records and accounts. It is strongly advised that one must write immediately after the operations i.e buying and selling.

b) Importance of records keeping

To know when and where to produce and where to sell; Farmer make the best decision concerning the choice of enterprise to undertake; Know how much money has been spent on each farming activity; Determine total sales (output); Know trends of the business; Access financial assistance- loans acquisition; Know the creditors and how much is owed and vise versa, Improve productivity by lowering costs of production per unit of output and; Know the value of the business.

c) Type of farm records and accounts

These includes Farm plan; Crop and livestock production records, Breeding and seed variety records, Farm assets; Financial records mainly cash books, petty cash, cash analysis, profit and loss account, balance sheets, ledgers and journals, Labor records; Amounts of consumption other than for sale

Amount of post-harvest (produce) losses

d) Financial records

Financial records are records of what the farmer spends and receives in his or her farm business. There are many financial records that the farmer can keep but the simplest of them is the cash book. Others include

- ✓ Ledger Book
- ✓ Receipt Book for any payments received
- ✓ Receipts for any expenses made
- ✓ Payment Vouchers
- ✓ Petty Cash Book/Vouchers
- ✓ Bank statements
- ✓ Assets and their value (appreciation and depreciation)
- ✓ Cash at hand
- ✓ Business agreements and or contracts,
- ✓ Payments
- ✓ Cash book

It summarizes what the farmer/manager spends and receives (Cash in or cash out) in his farm business

Anytime a farmer buys or sells an item for the business he or she enters the transaction in the cash book

A cash book has two pages. One page marked sales and receipts and the other page marked purchases and expenses.

Table 2: Daily Record sheet - cash book

SALES AND RECEIPTS				**PURCHASES AND EXPENSES**			
DATE	DETAILS	Ksh	cts	DATE	DETAILS	Ksh	cts
2.1.2004	Sold maize	5,000	00	3.2.2004	Hired tractor	2,500	00
30.1.2004	Received milk payment	10,000	00	28.2.2004	Bought fertilizer for maize	2,600	00
					Bought maize seed	2,000	00

I. CROP RECORD

The farm map is the beginning of farm records

Should show:

- -Partitions for various enterprises
- - Unused land
- - Pathways
- - Homestead

Other development

a. Examples crop records

Crop………………………………………………..

Area……………………………………………….

Date of planting	Inputs used		Date of harvest	Yields	Value	Remarks
	Amounts	Value				
12th march	Seed Maize-10kg. Beans-25kg. DAP-1 bag.					Poor weather

II. Labor records

Activity	No of man - days	Total cost	Remarks
Land preparation			
Planting			
1st weeding			
2nd weeding			

Once the above table has been filled, then Gross Margins for those particular crops can be calculated and the final decision can be made as pertaining to that particular crop.

III. Summary of Inputs/Output

Input costs	Quantity/Units	Unit Cost/earning Kshs	Total Cost/Earnings kshs	Remarks
Training and information costs				
Soil and water sampling and analysis				
Land preparation (per acre)				Land opening, harrowing, 2nd ploughing etc
Soil Conditioners (kg/litres)				
Planting Seed KG				
Fertilizers Planting Kg				
Planting man-days (MD)				
Fertilizers Topdressing Kg				
1st weeding MD				
2nd weeding MD				
Pest control MD				
Pesticides Litres				
Harvesting MD				
Transport Per Trip				
Shelling and Winnowing MD				
Drying				
Storage dusts				
Store/Silo (depreciation)				
Transport to market / Collection centre (trips)				
Advisory services (no of visits				
Credit (interest per month)				
Insurance (premiums)				
Total Expenses kshs				

Output

Output	Quantity/Units	Remarks
Production per in Kg per acre		
Unit price		
Gross Earnings		
Total Expenses kshs		
Profit in Kshs		
Fixed costs		
Net Earnings		

Av Monthly Income		

e) Costing of Farm Produce

Cost Statement for each product

DIRECT COSTS	No of Units	Unit cost kShs	Total Cost Kshs
Direct Materials Inputs			
Direct Labour			
Farm workers			
Direct Expenses Transport to get inputs			
Sub-Total Direct Costs (Prime Costs)			

INDIRECT COSTS

INDIRECT COSTS	No of Units	Unit cost kShs	Total Cost Kshs
Rent (eg lease of land)			
Salaries/wages			
Licenses			
Repairs/service of machinery & Equipment			
Transport			
Interest on Loan			
Water			
Fuel / Electricity			
Bad debts			
Others			
Sub -Total Indirect Costs (Overheads)			

TOTAL PRODUCTION COST = prime costs (direct costs) + overheads (indirect costs) Add profit (…... %) = selling price

Conclusion

Calculating your costs accurately and determining an appropriate price for it will allow you to cover all your costs and sell in sufficient quantities to make a profit and succeed in your business.

Appendices

Appendix I: Constraints Noted in Maize farming in areas.

Maize is a very important food crop but it faces a number of challenges in production. Some of these challenges include:

k) Erratic rainfall - poor distribution of rainfall results in total or partial crop failures,

l) Pests such as Stalk borer causes damage ranging from 60- 95% yet many farmers take no control measures,

m) Inadequate skills in diseases monitoring and pests scouting by farmers. Only 20% of farmers surveyed actual undertake regular scouting for pests and diseases This makes it more difficult to execute prevention and eradication measures at the right time.

n) Crops damage by wildlife was also raised by farmers that neighbor forested areas.

o) Diseases especially those with significant damage on leaves (Chlorosis and necrosis), stem (rot and lodging) and cobs (rotten and moldy).

p) Soil pests (termites, nematodes, cutworm, chaffer grubs), Rodents and Birds damage especially in maize from planting to cob formation,

q) Nutritional deficiency symptoms i.e. phosphorus especially in lower Eastern region were high,

r) Inadequate appropriate storage capacity at farm level, Lack of appropriate storage capacity poor post harvest management at small scale farm level results in heavy losses to harvested grains.

s) Continuous land subdivision into uneconomical units is affecting seed maize and commercial production.

t) Competition from cash crops such as sugarcane, tea, wheat, etc in major maize growing zones is reducing hectare available.

References:

a) De Lima, C.P.F. 1979. The assessment of losses due to insects and rodents in maize stored for subsistence in Kenya. *Trop. stored Prod. Inf.*, 38, 21-26.

b) Fielder, L. A. 1988. Rodent Problems in Africa. FAO Technical Publication, Pages 35-65.

c) Gratz, N. G. 1997. The burden of rodent-borne diseases in Africa south of Sahara. *Belgium Journal of Zoology*, 127: 71-84.

d) Mwanjabe, P. S and H. Leirs. 1997. An early warning system for IPM-based rodent Control in small holder farming systems in Tanzania. *Belgium Journal of Zoology*, 127: 49-58.

e) Taylor, K. D. 1962. Report on a two-day visit to the northern region of Tanganyika. *MAFF unpublished report,* 1-10.

f) Taylor, K. D. 1968. An outbreak of rats in agricultural areas of Kenya in 1962. *East African Agriculture and Forestry Journal*, 34: 66-77.

g) Taylor, K. D and G. M. Green. 1976. The influence of rainfall on diet and reproduction in four African rodent species. *Journal of Zoology, London*, 174: 453-471.